mathongo

JEE MAIN CRASH COURSE

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Designed by
Math Maestro

Anup Sir

JEE Mains 2020 Jan Chapter wise Question Bank

Permutation and Combination

Q1

The six digit numbers that can be formed using digits 1, 3, 5, 7, 9 such that each digit is used at least once.

7^{th} Jan Morning

Sol

1800

1, 3, 5, 7, 9

For digit to repeat we have ${}^5C_1$ choice

And six digits can be arrange in $\frac{\lfloor 6}{\lfloor 2}$ ways.

Hence total such numbers = $\frac{5\lfloor 6}{\lfloor 2}$

02

In a bag there are 5 red balls, 3 white balls and 4 black balls. Four balls are drawn from the bag. Find the number of ways of in which at most 3 red balls are selected

एक थेले में 5 लाल गेंदें, 3 सफेद गेंदें, और 4 काली गेंदें है थेले से 4 गेंदें निकाली जाती है अधिक से अधिक 3 लाल गेंदें चुने जाने के क्रमचयों की संख्या है।

(1) 450 (2) 360 (3) 490 (4) 510

8^{th} Jan Morning

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(3)

0 Red, 1Red, 2 Red, 3 Red

Number of ways क्रमचयों की संख्या = ${}^7C_4 + {}^5C_1.{}^7C_3 + {}^5C_2.{}^7C_2 + {}^5C_3.{}^7C_1 = 35 + 175 + 210 + 70 = 490$

03

The number of four letter words that can be made from the letters of word "EXAMINATION" is

8^{th} Jan Evening

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2454

EXAMINATION

2N, 2A, 2I, E, X, M, T, O

Case I All are different so $^8P_4 = \frac{8!}{4!} = 8.7.6.5 = 1680$

Case II 2 same and 2 different so $^3C_1 . ^7C_2 . \frac{4!}{2!} = 3.21.12 = 756$

Case III 2 same and 2 same so $^3C_2 . \frac{4!}{2!.2!} = 3.6 = 18$

$\therefore$ Total = 1680 + 756 + 18 = 2454

04

If number of 5 digit numbers which can be formed without repeating any digit while tenth place of all of the numbers must be 2 is 336 k find value of k

बिना किसी अंक की पुनरावृत्ति के बनने वाले 5 अंको की संख्याओं की संख्या जबकि दहाई के स्थान पर सभी संख्याओं में 2 आता हो 336 k है तो k का मान है–

(1) 8 (2) 7 (3) 6 (4) 5

9^{th} Jan Morning

Sol

(1)

			2	

Number of numbers संख्याओं की संख्या = $8 \times 8 \times 7 \times 6 = 2688 = 336k \Rightarrow k = 8$